**Barangay Relief Pack Optimizer**

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## 1. Application Overview

The Barangay Relief Pack Optimizer is a desktop application built in Python using Tkinter. It is designed to help barangay disaster response officers make two critical decisions quickly and accurately after a typhoon: what relief goods to pack into each bag, and which families should receive bags first when supply is limited.

The application has three screens accessible from the left sidebar:

Inventory Setup - enter relief goods and run the bag optimizer

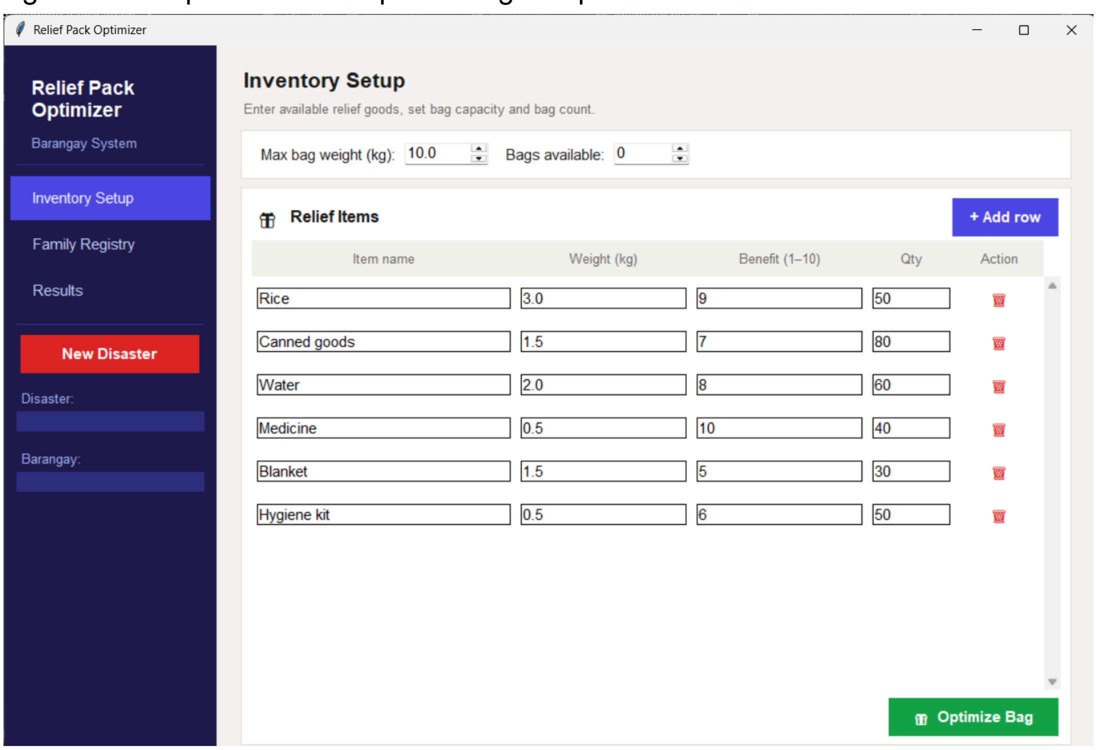
Family Registry - register affected families and generate the assignment

Results - view the optimized bag and ranked assignment report

## 2. Screen 1: Inventory Setup

### 2.1 Purpose

This screen is where the officer enters the available relief goods, sets the maximum bag weight, and specifies how many bags are available. Clicking Optimize Bag runs the 0/1 Knapsack DP algorithm and produces the optimal bag composition.




### 2.2 Input Fields

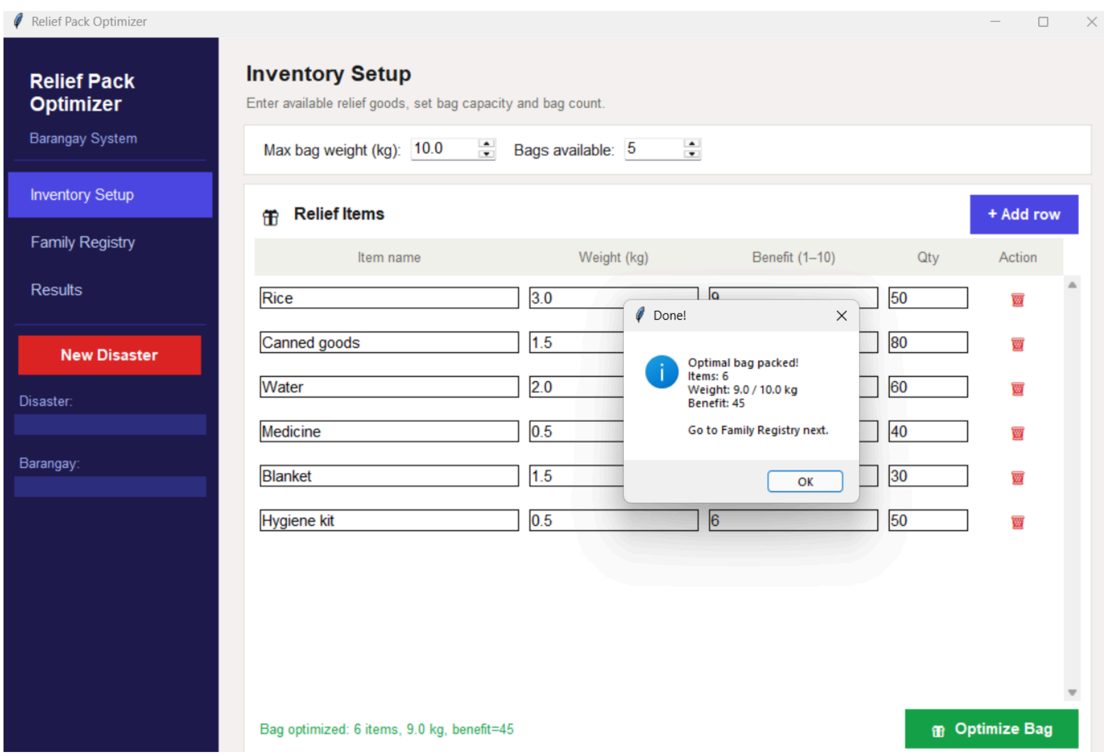
**Max bag weight (kg):** The maximum weight a single relief bag can hold. Entered using a numeric spinner.
**Bags available:** The total number of relief bags the barangay can distribute.
Relief Items Table: Each row represents one item type with four fields: Item name, Weight (kg), Benefit score (1–10), and Quantity available.

Default items pre-loaded into the table:

- Rice - 3.0 kg, benefit 9
- Canned goods - 1.5 kg, benefit 7
- Water - 2.0 kg, benefit 8
- Medicine - 0.5 kg, benefit 10
- Blanket - 1.5 kg, benefit 5
- Hygiene kit - 0.5 kg, benefit 6

### 2.3 How to Use

1. Set the maximum bag weight using the spinner (e.g., 10 kg).
2. Set the number of bags available.
3. Review or edit the relief items table. Use + Add row to add new items. Click the trash icon to remove a row.
4. Click Optimize Bag. The system runs the DP algorithm and displays a confirmation showing the number of items selected, total weight, and total benefit score.



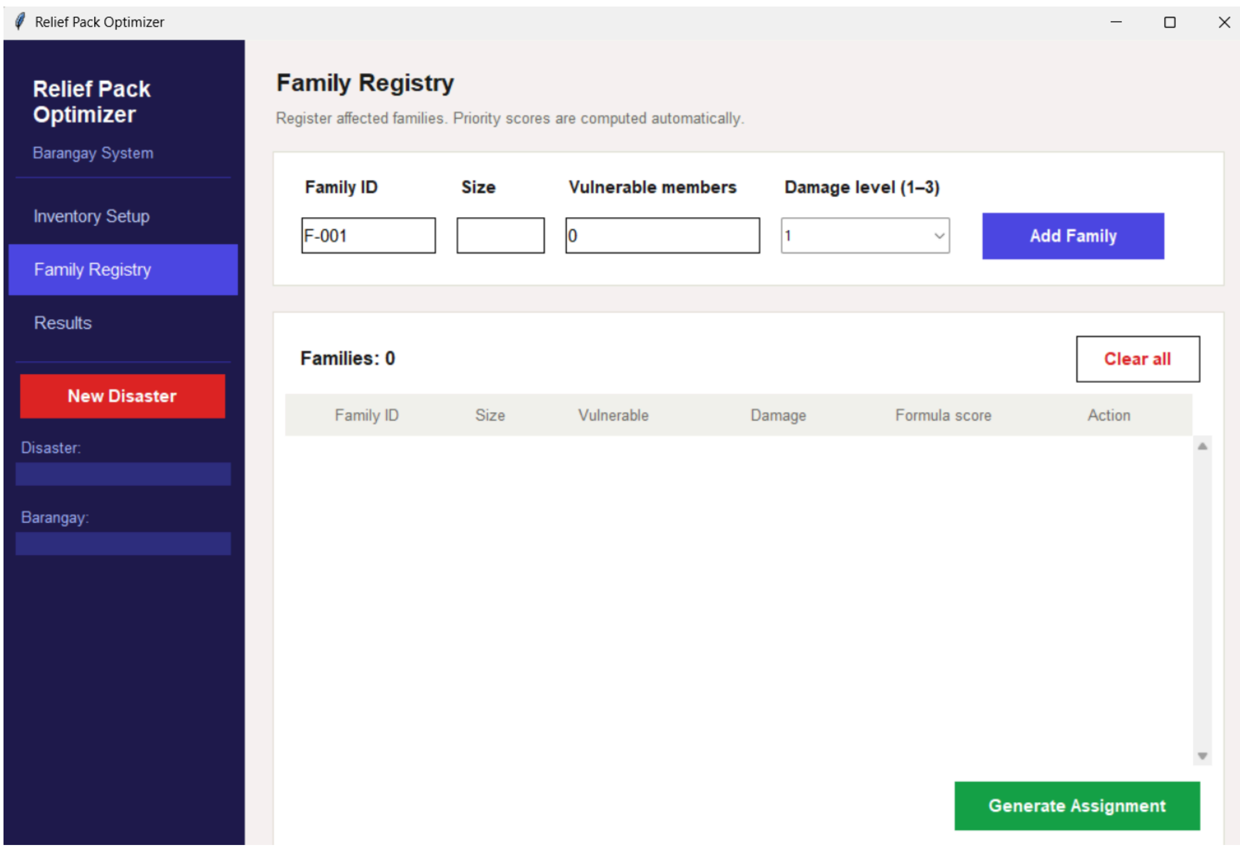

### 2.4 What Happens Internally

The Optimize Bag button triggers the 0/1 Knapsack DP algorithm in knapsack.py. Item weights are scaled to 0.1 kg precision to allow integer DP table indexing. A 2D table dp[i][w] is filled bottom-up where each cell stores the maximum benefit achievable using the first i items with capacity w. The algorithm then backtracks through the table to identify exactly which items were selected.

## 3. Screen 2: Family Registry

### 3.1 Purpose

This screen is where the officer registers each affected family. Each family is automatically assigned a priority score based on the formula below. Clicking Generate Assignment runs the greedy ranking and bag assignment.




### 3.2 Priority Score Formula

priority_score = (family_size × 2) + (vulnerable_count × 5) + (damage_level × 3)

**family_size:** Total number of household members

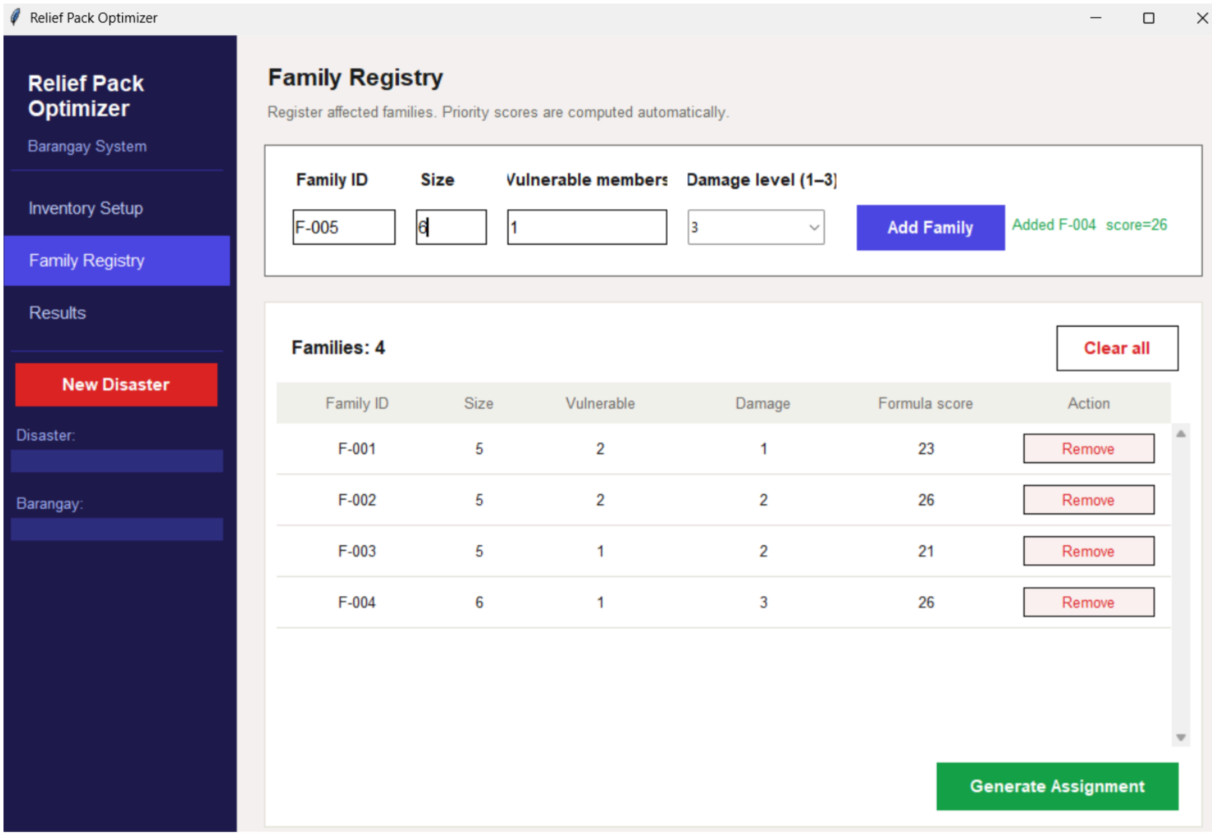
**vulnerable_count:** Number of elderly, infants, or pregnant members (weighted highest at 5 - greatest health risk)

**damage_level:** 1 = minor damage, 2 = partial loss, 3 = total loss

Tiebreaker rule: if two families have equal scores, the larger household is ranked first. If still equal, the family registered earlier takes precedence.

### 3.3 Input Fields

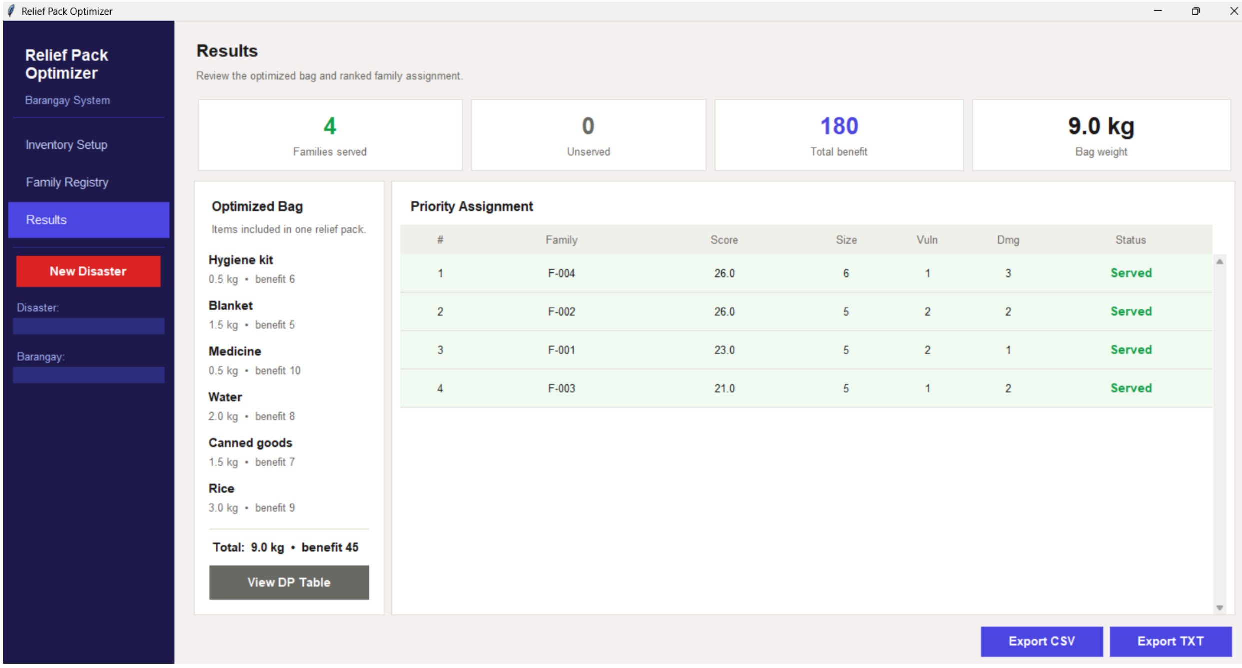
**Family ID:** A unique identifier for the household (e.g., F-001)
**Size**: Total number of household members (1–20)
**Vulnerable members:** Count of elderly, infants, or pregnant members
**Damage level:** Selected from dropdown: 1 (minor), 2 (partial), 3 (total loss)




## 4. Screen 3: Results

### 4.1 Purpose

This screen displays the complete output of both algorithms, the optimized bag manifest on the left and the ranked family assignment table on the right. A summary strip at the top shows key statistics at a glance.




### 4.2 Summary Strip

Four cards at the top of the screen display:

- Families served - how many families received a bag
- Unserved - how many families did not receive a bag due to supply shortage
- Total benefit - total benefit score delivered across all served families
- Bag weight - the total weight of one optimized bag

### 4.3 Optimized Bag Manifest (Left Panel)

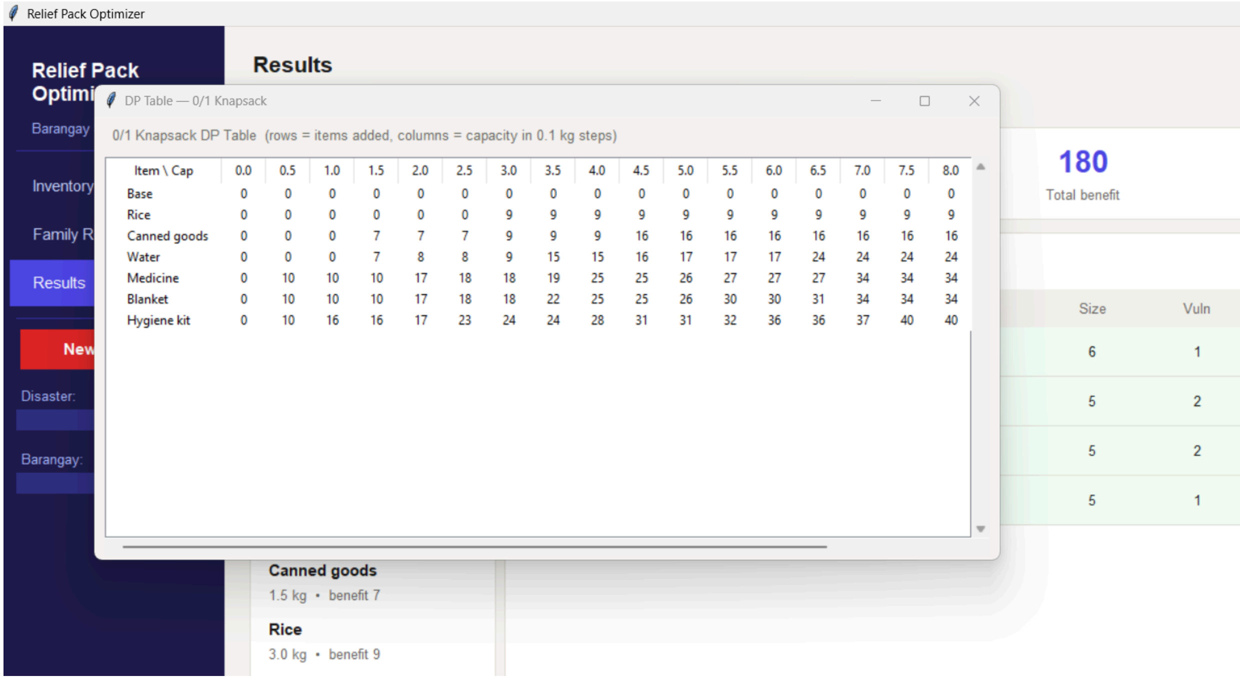
Lists every item selected by the DP algorithm for one relief bag. Each item shows its name, weight, and benefit score. The total weight and total benefit are displayed at the bottom.

### 4.4 Priority Assignment Table (Right Panel)

Lists all registered families ranked by priority score from highest to lowest. Each row shows the family ID, final score, household size, vulnerable count, damage level, and status (Served or No Supply). Served families are highlighted in green, unserved in red.

### 4.5 View DP Table

The View DP Table button opens a separate window displaying the full 0/1 Knapsack DP table used in Module 1. Rows represent items added one by one. Columns represent bag capacity shown in 0.5 kg steps for readability. Internally the algorithm works at 0.1 kg precision, so the full table has more columns than displayed. Each cell shows the maximum benefit achievable at that item count and capacity. This is provided for transparency and academic verification of the algorithm.

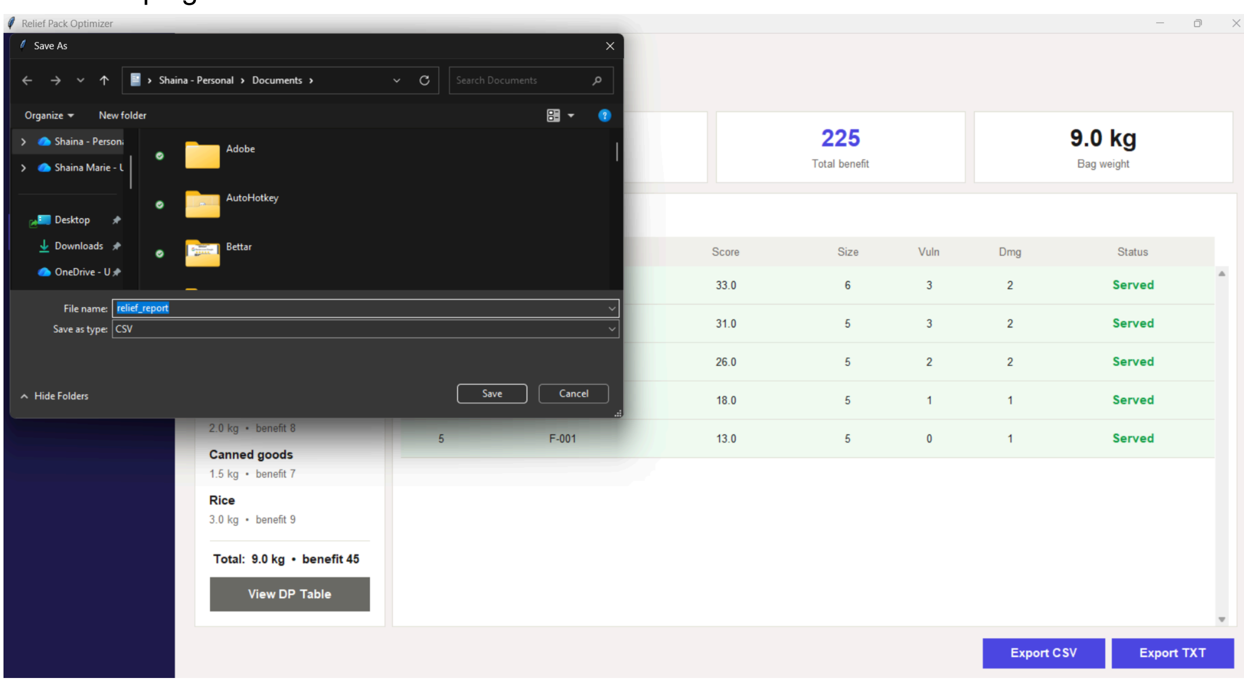


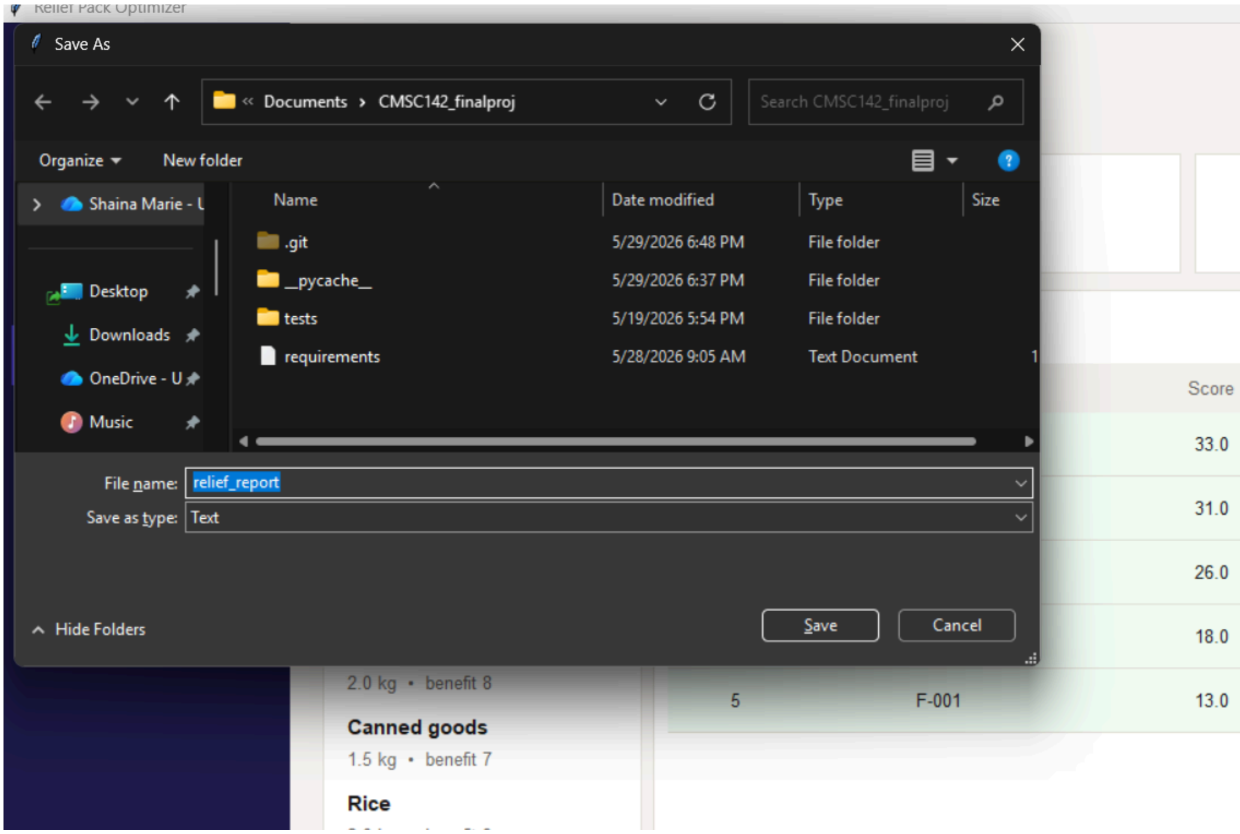

| Item \ Cap | 0.0 | 0.5 | 1.0 | 1.5 | 2.0 | 2.5 | 3.0 | 3.5 | 4.0 | 4.5 | 5.0 | 5.5 | 6.0 | 6.5 | 7.0 | 7.5 | 8.0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Base | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Rice | 0 | 0 | 0 | 0 | 0 | 0 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 |
| Canned goods | 0 | 0 | 0 | 7 | 7 | 7 | 9 | 9 | 9 | 16 | 16 | 16 | 16 | 16 | 16 | 16 | 16 |
| Water | 0 | 0 | 0 | 7 | 8 | 8 | 9 | 15 | 15 | 16 | 17 | 17 | 17 | 24 | 24 | 24 | 24 |
| Medicine | 0 | 10 | 10 | 10 | 17 | 18 | 18 | 19 | 25 | 25 | 26 | 27 | 27 | 27 | 34 | 34 | 34 |
| Blanket | 0 | 10 | 10 | 10 | 17 | 18 | 18 | 22 | 25 | 25 | 26 | 30 | 30 | 31 | 34 | 34 | 34 |
| Hygiene kit | 0 | 10 | 16 | 16 | 17 | 23 | 24 | 24 | 28 | 31 | 31 | 32 | 36 | 36 | 37 | 40 | 40 |

## 4.6 Export Options

Two export buttons are available at the bottom of the Results screen:

- Export TXT - generates a plain text report of the bag manifest and assignment list, suitable for printing and field use
- Export CSV - generates a spreadsheet-compatible file of the assignment table for record keeping




Relief Pack Optimizer
Save As
Documents › CMSC142_finalproj
Search CMSC142_finalproj
Organize
New folder
Shaina Marie - U
Desktop
Downloads
OneDrive - U
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Name
Date modified
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5/29/2026 6:48 PM
File folder
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requirements
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Text
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33.0
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18.0
2.0 kg • benefit 8
5
F-001
13.0
Canned goods
1.5 kg • benefit 7
Rice